ENGINEERING MATHEMATICS-II

Kavyashree
Department of Science and Humanities

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FOURIER TRANSFORMS

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Fourier Cosine Transform:

Definition:

The Fourier Cosine Transform for the function f(t) is defined by

$$\mathcal{F}_c[f(t)] = \boxed{F_c(\omega) = \int_0^\infty f(t)\cos\omega t\,dt}$$

Definition:

The Inverse Fourier Cosine Transform for the function $F_c(\omega)$ is defined by

$$\boxed{f(t) = \frac{2}{\pi}\int_0^\infty F_c(\omega)\cos\omega t\,d\omega}$$

Fourier Cosine Transforms:

Problem 1: Find Fourier Cosine Transform of e^{-at} $(a > 0)$ and hence evaluate

$$(i) \int_0^\infty \frac{\cos \lambda x}{x^2 + a^2} dx \qquad ii) \mathcal{F}_c(te^{-at})]$$

Solution: We have,

$$\mathcal{F}_c[f(t)] = \int_0^\infty f(t) \cos \omega t dt = F_c(\omega)$$

$$\mathcal{F}_c[e^{-at}] = \int_0^\infty e^{-at} \cos \omega t dt$$

$$= \left[\frac{e^{-at}}{a^2+\omega^2} \{-a \cos \omega t + \omega \sin \omega t\} \right]_0^\infty$$

$$\boxed{F_c(\omega) = \frac{a}{a^2+\omega^2}}$$

Fourier Cosine Transforms:

$$(i)\ f(t) = \frac{2}{\pi}\int_0^\infty F_c(\omega)\cos\omega t\, d\omega$$

$$\Rightarrow e^{-at} = \frac{2}{\pi}\int_0^\infty \frac{a}{a^2+\omega^2}\cos\omega t\, d\omega$$

$$t = \lambda \Rightarrow e^{-a\lambda}\frac{\pi}{2a} = \int_0^\infty \frac{\cos\omega\lambda}{a^2+\omega^2}\, d\omega;$$

$$\omega \to x \Rightarrow \int_0^\infty \frac{\cos x\lambda}{a^2+x^2}\, dx = \frac{\pi}{2a}e^{-a\lambda}$$

Fourier Cosine Transforms:

$$ii)\ \mathcal{F}_c[e^{-at}] = \frac{a}{a^2+\omega^2} = F_c(\omega)$$

$$\frac{d}{da}\{\mathcal{F}_c[e^{-at}]\} = \frac{d}{da}\left(\frac{a}{a^2+\omega^2}\right) \Rightarrow \frac{d}{da}\left(\int_0^\infty e^{-at}\cos\omega t\,dt\right) = \frac{d}{da}\left(\frac{a}{a^2+\omega^2}\right)$$

$$\Rightarrow \left(\int_0^\infty \frac{\partial}{\partial a}\left\{e^{-at}\cos\omega t\,dt\right\}\right) = \frac{a^2+\omega^2-a(2a)}{\left(a^2+\omega^2\right)^2}$$

$$\Rightarrow \int_0^\infty -te^{-at}\cos\omega t\,dt = \frac{\omega^2-a^2}{\left(a^2+\omega^2\right)^2}$$

$$\Rightarrow F_c\left[te^{-at}\right] = \frac{a^2-\omega^2}{\left(a^2+\omega^2\right)^2}$$

Fourier cosine Transforms:

Problem 2: Find Fourier Cosine Transform of $t^{n-1}\ (n \geq 2)$ and hence find $\mathcal{F}_c\left[\frac{1}{\sqrt{t}}\right]$

Solution: $$\mathcal{F}_c[f(t)] = \int_0^\infty f(t)\cos\omega t\,dt$$

$$\text{i.e.,}\ \mathcal{F}_c[t^{n-1}] = \int_0^\infty t^{n-1}\cos\omega t\,dt$$

$$= \operatorname{Re}\int_0^\infty t^{n-1}e^{-i\omega t}\,dt = \operatorname{Re}\frac{\Gamma(n)}{(i\omega)^n}$$

$$\left(\because \int_0^\infty x^{n-1}e^{-ax}\,dx = \frac{\Gamma(n)}{(a)^n}\right)$$

$$= \operatorname{Re}\frac{\Gamma(n)}{i^n\omega^n} \qquad = \operatorname{Re}\frac{\Gamma(n)}{\omega^n}(-i)^n$$

$$= \operatorname{Re}\frac{\Gamma(n)}{\omega^n}\left(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}\right)^n$$

Fourier cosine Transforms:

$$\mathcal{F}_c[t^{n-1}] = \frac{\Gamma(n)}{\omega^n}\cos\frac{n\pi}{2} = \frac{n!}{\omega^n}\cos\frac{n\pi}{2}$$

$$n = \frac{1}{2} \Rightarrow \mathcal{F}_c\left[\frac{1}{\sqrt{t}}\right] = \frac{\Gamma\left(\frac{1}{2}\right)}{\omega^{\frac{1}{2}}}\cos\frac{\pi}{4}$$

$$= \frac{\sqrt{\pi}}{\sqrt{\omega}}\frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{\pi}}{\sqrt{2}}\frac{1}{\sqrt{\omega}}$$

Fourier Cosine Transforms:

Problem 3: Solve the integral equation $\int_0^\infty f(\theta)\cos\alpha\theta\, d\theta = \begin{cases} 1-\alpha, & 0\le\alpha\le 1 \\ 0, & otherwise \end{cases}$

and hence evaluate $\int_0^\infty \frac{\sin^2 t}{t^2}\, dt$

Solution:

$$F_c(\alpha) = \begin{cases} 1-\alpha, & 0\le\alpha\le 1 \\ 0, & otherwise \end{cases} \qquad f(\theta) = ?$$

$$f(\theta) = \frac{2}{\pi}\int_0^\infty F_c(\alpha)\cos\alpha\theta\, d\alpha = \frac{2}{\pi}\int_0^1 (1-\alpha)\cos\alpha\theta\, d\alpha$$

$$= \frac{2}{\pi}\left[(1-\alpha)\frac{\sin\alpha\theta}{\theta} - \frac{(-\cos\alpha\theta)}{\theta^2}(-1)\right]_0^1$$

$$f(\theta) = \frac{2}{\pi}\frac{-(\cos\theta-1)}{\theta^2} = \frac{2}{\pi}\frac{(1-\cos\theta)}{\theta^2}$$

Fourier Cosine Transforms:

$$\int_0^\infty f(\theta)\cos\alpha\theta\, d\theta = \frac{2}{\pi}\int_0^\infty \frac{(1-\cos\theta)}{\theta^2}\cos\alpha\theta\, d\theta = F_c(\alpha);$$

$$\alpha = 0 \Rightarrow F_c(\alpha=0) = 1$$

$$\Rightarrow \frac{2}{\pi}\int_0^\infty \frac{(1-\cos\theta)}{\theta^2} d\theta = 1 \Rightarrow \int_0^\infty \frac{(1-\cos\theta)}{\theta^2} d\theta = \frac{\pi}{2}$$

$$Put\ \theta = 2t \Rightarrow d\theta = 2dt$$

$$\Rightarrow \int_0^\infty \frac{(1-\cos 2t)}{4t^2} 2dt = \frac{\pi}{2}$$

$$\boxed{\therefore \int_0^\infty \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}}$$

PES UNIVERSITY

THANK YOU

Kavyashree
Department of Science and Humanities
kavyashree@pes.edu